Ambidexterity and absorptive capacity in boundary-spanning managers: role of paradox mindset and learning goal orientation

Saurav Snehvrat
Department of Strategic Management, XLRI - Xavier School of Management, Jamshedpur, India
Sanjay Chaudhary
Department of Strategic Management, Jindal Global Business School, O.P. Jindal Global University, Sonipat, India, and
Siddharth Gaurav Majhi
IT and Computational Systems Area, Indian Institute of Management Amritsar, Amritsar, India

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Abstract

Purpose – Boundary-spanning managers need to recognize, learn and implement external knowledge while balancing the conflicts emerging from new and existing knowledge. The authors' study explores how a paradox mindset (PM) and a learning focus [learning goal orientation (LGO)] promote two managerial capabilities: absorptive capacity (ACAP) and ambidexterity. The authors' study explores the inter-relationship between the mindsets and the capabilities required for innovative work behavior.

Design/methodology/approach – The authors use survey data from 113 technology/product managers employed in boundary-spanning roles in a large Indian automotive equipment manufacturing firm. Partial least squares structural equation modeling (PLS-SEM) analysis and bootstrapping (using PROCESS MACRO) are used to test for direct and mediation effects respectively.

Findings – Both PM and LGO are found to affect individual ambidexterity (IA) via the mediation of individual absorptive capacity (IACAP). While IACAP partially mediates the relationship between PM and IA, there is full mediation in the case of LGO.

Research limitations/implications – The authors focus on a sample of managers from a single, large Indian automotive firm. Although single case studies can help provide novel conceptual insights and to test theoretical relationships, future research needs to confirm the authors' findings in different types of firms.

Practical implications – This study shows how a learning orientation and the ability to be energized from conflicts help boundary-spanning managers produce innovative outcomes.

Originality/value – The authors reveal fresh insights on how both ACAP and ambidexterity share the focus on learning and paradox management. The authors explicate how LGO and PM uniquely impact the critical capabilities of IACAP and IA for boundary-spanning managers.

Keywords Individual ambidexterity, Individual absorptive capacity, Paradox mindset, Boundary spanners, Learning goal orientation, Innovative work behavior

Paper type Research paper

1. Introduction

In the present-day business environment, firms seeking to pursue innovation must actively engage with a wide variety of fragmented yet useful information residing outside their boundaries (Ritala and Stefan, 2021). This is especially relevant for emerging market firms that typically lack access to advanced resources, knowledge and capabilities (Badir *et al.*, 2020; Bogers *et al.*, 2019). Thus, the managerial ability to deal with external knowledge and internal resource constraints becomes increasingly important in emerging market firms.

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Typically, boundary-spanning managers are more likely to engage with external knowledge regularly. These individuals transact with the relevant individuals outside the organization and play a key role in facilitating and managing inter-organizational transactions (Fleming and Waguespack, 2007; Richter *et al.*, 2006). Such boundary spanners benefit from possessing individual absorptive capacity (IACAP), which allows them to identify, assimilate, transform and implement external knowledge to benefit organizational and individual outcomes such as innovative work behavior (Lowik *et al.*, 2017; Majhi *et al.*, 2020). Additionally, boundary spanners are often confronted with the need to balance conflicts emanating from the need to simultaneously pursue activities related to exploring new knowledge and exploiting existing knowledge (Mom *et al.*, 2007). The ability to effectively balance and deal with exploratory and exploitative learning activities is called individual ambidexterity (IA) (Mom *et al.*, 2009). Since IA is an important managerial capability, prior academic research has focused extensively on its key antecedents as well as its impact on key individual outcomes such as innovative work behavior (Mom *et al.*, 2019; Rosing and Zacher, 2017).

On the ground, individual employees implement organizational open innovation strategies into practice. On the one hand, firm employees, especially boundary spanners, can act as knowledge brokers by recognizing, assimilating, transforming and exploiting new knowledge/technologies for the firm (Felin and Hesterly, 2007; Grant, 1997). On the other hand, employees can also act as a hindrance to the adoption of open innovation activities by exhibiting biases (not-invented-here and not-sold-here) against them (Chesbrough, 2006; Lichtenthaler, 2011).

Our study utilizes knowledge and learning perspective by focusing on two related yet distinct concepts of IACAP and IA. A group of studies have analyzed the antecedents of IA (Ajayi *et al.*, 2017; Asif, 2017; Kao and Chen, 2016; Torres *et al.*, 2015) and IACAP (Chuang *et al.*, 2016; Ince *et al.*, 2022) separately. However, there have been calls for more research on identifying key antecedents of these critical managerial capabilities and their interplay, given their importance in an open innovation-based and knowledge-intensive business environment (Lowik *et al.*, 2017; Schnellbächer *et al.*, 2019; Wang and Gibbons, 2021; Xiang *et al.*, 2019). Given that these capabilities are deeply embedded in knowledge and learning, learning goal orientation (LGO) can be one such key antecedent (Islam *et al.*, 2021; Paparoidamis, 2005). Similarly, due to the tensions inherent in ambidextrous behavior (Venugopal *et al.*, 2018), a paradox mindset (PM) has been argued as a potentially important antecedent (Knight and Harvey, 2015; Tabesh and Vera, 2020). The discussion on these aspects has been missing in the extant academic literature. To address this gap, in this paper, we seek to extend the current understanding of the antecedents of IACAP and IA by examining how LGO and PM affect them while considering the interplay between the IACAP and IA.

LGO refers to the desire to improve one's competence in tasks (Annosi *et al.*, 2020; Farr *et al.*, 1993; Yokoyama and Miwa, 2021). High LGO motivates individuals to expend effort beyond current tasks and knowledge and also focus on learning and knowledge acquisition to gain the ability to execute tasks in the future (Button *et al.*, 1996; Heyman and Dweck, 1992). With both IACAP and IA being important managerial capabilities and both drawing from common roots in knowledge and learning-based literature (Cohen and Levinthal, 1990; March, 1991), LGO can have an effect on these capabilities. But this aspect has attracted limited research attention (Kauppila and Tempelaar, 2016; Torres *et al.*, 2015). Thus, the first question driving this study is as follows:

RQ1. What is the impact of LGO on the learning-based capabilities of IA and IACAP in boundary-spanning settings?

Paradox theory has emerged as a critical theoretical lens to examine issues pertaining to conflicts (Schad *et al.*, 2019). Recently, researchers have introduced the concept of PM to adopt a micro-foundational view on paradoxes (Miron-Spektor *et al.*, 2018). Since both IA and IACAP consist of multiple, inter-related activities which are inherently in conflict with each other (Martin *et al.*, 2019; Rafique *et al.*, 2018), PM can be an effective mechanism to deal with such conflicts in the context of these managerial capabilities. Miron-Spektor *et al.* (2018) explicitly ask for studies that "test . . . claims that a paradox mindset fosters ambidexterity and learning" (p. 28). Some studies have explored the impact of a PM on employee behavior (Birkinshaw and Gibson, 2004; Carmeli and Halevi, 2009; Miron-Spektor *et al.*, 2018). However, the impact of PM on managerial capabilities such as IA and IACAP has not been examined despite some investigation at the organizational level (Beletskiy and Fey, 2020). Hence, the second question driving this study is as follows:

RQ2. What is the impact of the PM on IA and ACAP in boundary-spanning settings?

This paper contributes to the extant academic literature by providing novel insights regarding the inter-relationship between distinct capabilities of boundary-spanning managers (IA and IACAP). This study also provides a holistic understanding of the inter-relationships between learning and conflict-related mindsets and knowledge-based managerial capabilities. By demonstrating the critical mediating role of IACAP between PM/LGO and IA, our study brings out important inter-connections between two distinct knowledge-related capabilities in the boundary-spanning context.

2. Theoretical background and hypotheses development

2.1 Individual ambidexterity (IA)

Ambidexterity refers to an ability to balance the trade-offs between exploration and exploitation (Raisch *et al.*, 2009). Its importance is reflected in the failure of organizations such as Kodak and Boeing due to lack of ambidexterity, and the success of organizations like USA Today and Ciba Vision, who managed to develop ambidextrous capabilities (O'Reilly and Tushman, 2004). While exploration entails experimentation with new knowledge, technologies, processes, routines and markets (characterized by terms like search, variation, risk-taking, experimentation, play, flexibility, discovery and innovation), exploitation includes incremental improvements on existing knowledge and dependence on previous experience regarding technology, systems, processes and routines (characterized by terms like refinement, choice, production, efficiency, selection, implementation and execution) (March, 1991). Although ambidexterity has received substantial academic attention at the organizational level (O'Reilly and Tushman, 2013), attention on the individual-level manifestation of ambidexterity has been relatively limited (Bidmon and Boe-Lillegraven, 2020; Christofi *et al.*, 2021; Kauppila and Tempelaar, 2016; Papachroni and Heracleous, 2020; Rosing and Zacher, 2017). This is despite the realization from prior research that organizational or unit-level ambidexterity has its origins in managerial exploration and exploitation activities to a large extent (Mom *et al.*, 2007, 2019; Rosing and Zacher, 2017; Tempelaar and Rosenkranz, 2019).

IA is conceptualized as the capability of managers to simultaneously demonstrate exploration and exploitation of knowledge (Mom *et al.*, 2007). Exploration requires managers to adopt a long-term perspective and challenge well-established beliefs and thought processes (Rivkin and Siggelkow, 2003). On the other hand, exploitation entails refinement or building on their existing knowledge about processes, systems, processes and technologies with an efficient short-term approach (Mom *et al.*, 2015). Thus, IA is the behavioral capacity to effectively balance two conflicting elements of exploration and exploitation (Kauppila and Tempelaar, 2016). Individuals, generally, tend to fall into exploitation cycles and hence factors that push individuals toward exploration without severely affecting their exploitation ability are good candidates as the antecedents of IA (Levinthal and March, 1993). Although various antecedents of IA have been identified that span areas like organizational learning, organizational architecture, general self-efficacy, intrinsic motivation, psychological contract fulfillment, knowledge inflows, etc. (Ajayi *et al.*, 2017; Kao and Chen, 2016; Keller and Weibler, 2015; Mom *et al.*, 2019; Schnellbächer *et al.*, 2019; Tempelaar and Rosenkranz, 2019; Zhang *et al.*, 2019), in this study we focus on two important constituents of IA.

As an academic concept, ambidexterity has its roots in organizational learning and knowledge management literature (March, 1991; O'Reilly and Tushman, 2004; Raisch *et al.*, 2009). Also a recurring and fundamental theme in studies of ambidexterity is the presence, nature and management of conflicts emerging between exploration and exploitation. Hence, an individual's disposition toward learning/knowledge as well as inter-connected and persistent conflicts is indispensable antecedents of IA. Our study captures these two antecedents through the constructs of LGO and PM. Further, even with the presence of a tendency/disposition toward learning and comfort with conflicts, an individual still needs the ability to recognize, understand and assimilate new knowledge presented to his/her. This ability is captured in the study via the construct of IACAP.

2.2 Paradoxes and paradox mindset (PM)

Paradoxes are defined as "contradictory yet interrelated elements that exist simultaneously and persist over time" and can be classified into four categories – learning, organizing, belonging and performing (Smith and Lewis, 2011, p. 382). Paradox theory is concerned with how organizations and managers can cope positively with persistent conflicts to ignite positive and innovative outcomes (Putnam *et al.*, 2016). Recent focus has also been on individuals' experiences with paradoxes within organizations, including paradoxical leadership of top management team (TMT) members and Chief Executive Officers (CEOs) (Zhang and Han, 2019). Examining the micro-foundations of organizational paradoxes, Miron-Spektor *et al.* (2018) define PM as "the extent to which one is accepting of and energized by tensions" and show how this can help employees and managers create positive outcomes when confronted with conflicting and interrelated tensions. Individuals possessing a PM feel comfortable with contradictions and tensions and can proactively accept them (Hargrave and Van de Ven, 2017; Pan, 2021; Sleesman, 2019; Smith and Lewis, 2011).

PM has been found to add value in terms of in-role and innovation performance in varied contexts such as distributed IS development teams (McCarthy *et al.*, 2019), creative settings such as architecture (Gaim, 2018; Miron-Spektor *et al.*, 2011), digital workplace technologies (Schneider and Kokshagina, 2020), multi-occupation and multi-country samples (Miron-Spektor *et al.*, 2018), women leaders confronted with agency-communion tensions (Zheng *et al.*, 2018), work–family conflicts (Chen *et al.*, 2020), small and medium enterprises (Bin and Dashuai, 2020) and entrepreneurs (Klonek *et al.*, 2021).

2.3 PM and IA

Advancements in paradox theory have shed light on the inter-connectedness between inter-connected and persistent tensions/conflicts encountered within organizations, in general, and the ambidexterity concept, specifically (Knight and Harvey, 2015; Smith and Lewis, 2011; Tabesh and Vera, 2020). Both paradox theory and ambidexterity theory (exclusive focus on exploration-exploitation) focus on mechanisms that lead to the effective management of such conflicts. Recently, ambidexterity researchers have started to adopt a paradox theory lens to study the interplay between exploration and exploitation across levels (Asif, 2017; Klonek *et al.*, 2021; Knight and Paroutis, 2017; Smith and Tushman, 2005; Venugopal *et al.*, 2018). Paradox theory scholars, on the other hand, have detailed the concept of a PM that helps individuals improve innovation and in-role performance (Miron-Spektor *et al.*, 2018).

Both PM and IA have a set of common cognitive and contextual antecedents, such as cognitive flexibility, divergent thinking and organic work structures (Good and Michel, 2013; Klonek *et al.*, 2021; Laureiro-Martínez *et al.*, 2015). While there has been a tendency to equate PM and IA in some prior literature, scholars have recently distinguished the two by measuring both constructs for the same individuals (Klonek *et al.*, 2021). While PM refers to a tendency/disposition towards acceptance and comfort with tensions, IA refers to a demonstrated ability to balance the conflicting demands of exploration and exploitation. Martin *et al.* (2019) argue that the presence of conflicts serves as a micro-foundation of organizational ambidexterity and individual-level conflicts arise due to the presence of multiple roles, memberships and perspectives. Since paradoxical challenges are inherent in the management of ambidexterity at different units of analysis (Bidmon and Boe-Lillegraven, 2020; Miron-Spektor *et al.*, 2018), PM can be effective in managing the challenges associated with the pursuance of ambidextrous behavior by individuals, thus enhancing IA (Kauppila and Tempelaar, 2016). PM enables individuals to balance contradictory demands (such as those related to exploration and exploitation).

This relationship has been established both conceptually and empirically in prior literature (Andriopoulos and Lewis, 2009; Papachroni *et al.*, 2016; Smith, 2014; Smith and Tushman, 2005). Thus, instead of an either/or approach to managing the exploration–exploitation tensions in IA, PM helps foster a dynamic balance between the two to achieve a more synergistic outcome (Papachroni and Heracleous, 2020). By enhancing creativity and reducing the tendency to stick to established cognitive frames, PM allows the flexibility to balance exploration and exploitation (Hargrave and Van de Ven, 2017; Miron-Spektor *et al.*, 2011). PM also helps managers effectively deal with the resource constraints associated with the need to balance exploratory and exploitative activities (Beletskiy and Fey, 2020; Kauppila, 2010).

Hence, we hypothesize as follows:

H1. The PM of a manager enhances his or her IA.

2.4 Individual absorptive capacity (IACAP)

Defined as "the ability of a firm to recognize the value of new, external information, assimilate it, and apply it to commercial ends" (Cohen and Levinthal, 1990, p. 128), ACAP (ACAP) helps firms gain new knowledge from external sources in an attempt to adapt to fast-changing business environments (Seo *et al.*, 2015; Zahra and George, 2002). Although ACAP has received significant attention at the organizational level, there is also widespread agreement among researchers regarding the role of the individual manager in gaining and implementing external knowledge for the firm. Given that the capability of individuals to manage external knowledge flows is fundamental to organizational ACAP (Cohen and Levinthal, 1990; Lowik *et al.*, 2017; Matusik and Heeley, 2005; Volberda *et al.*, 2010), there has been a renewed focus on IACAP, defined as the ability of individuals to identify, assimilate and utilize new knowledge emanating from external sources (Enkel *et al.*, 2017).

IACAP comprises four distinct but sequential activities: identification/recognition of new knowledge, assimilation of the identified knowledge, a transformation of the assimilated knowledge and finally exploitation of the transformed knowledge (Enkel *et al.*, 2017; Hafkesbrink and Schroll, 2014). First, individuals must be able to recognize new knowledge opportunities (Enkel *et al.*, 2017; Zahra and George, 2002). Then, they must select and assimilate elements of the recognized knowledge that are most relevant (Enkel *et al.*, 2017; Zahra and George, 2002). After assimilation, multiple elements of the new knowledge need to be combined with the existing firm knowledge (Enkel *et al.*, 2017; Zahra and George, 2002). In this stage, potential conflicts between new and old knowledge, systems, technologies and processes must be overcome through the appropriate transformation of the newly acquired knowledge (Enkel *et al.*, 2017; Zahra and George, 2002). Finally, managers must be able to drive the implementation of the ideas based on the newly acquired knowledge within their firms (Enkel *et al.*, 2017). This stage requires managers to positively influence changes in systems, technologies, processes, structures and cultures within the firm (Zahra and George, 2002).

2.5 IACAP and IA

Absorptive capacity theory and ambidexterity theory have had their respective largely non-overlapping bodies of literature. However, there are obvious points of overlap between ACAP and ambidexterity. Both theories trace their origins to organizational learning and knowledge management literature (March, 1991; Zahra and George, 2002). Further, both concepts deal with the interplay between new and existing knowledge.

Since IA refers to the ability to balance exploratory (radical) and exploitative (incremental) innovation, it frequently requires a combination of external and internal knowledge to deal with potential resource constraints (Kauppila, 2010). IACAP deals with the identification and induction of external knowledge. Extant research points at the complementary role of internal and external knowledge in producing innovative outcomes (Faems *et al.*, 2010; West and Bogers, 2014). On the other hand, a few studies have pointed at the probability of limited search and variation because of heightened IACAP (Lichtenthaler, 2016; Majhi *et al.*, 2020). In such situations, managers must combine IACAP with IA to consistently achieve innovative outcomes. Additionally, some studies have explored the impact of the four activities (discussed earlier) inherent in ACAP with individual exploration and exploitation activities independently (Enkel *et al.*, 2017; Ince *et al.*, 2022; Khan *et al.*, 2019). In essence, the ability of managers to seek and assimilate external knowledge can help in balancing exploration and exploitation. Hence, we argue that IACAP is an important pre-condition for managers to achieve a balance between exploratory and exploitative innovation. Thus, we hypothesize as follows:

H2. The IACAP of a manager enhances his or her IA.

2.6 PM and IACAP

In the previous sections, we have explained how extant research suggests a relationship between PM and IA and how IACAP leads to increased exploration and exploitation. The construct of PM is defined as a tendency/disposition that helps individuals to value, accept and be energized from paradoxical conflicts. IACAP, on the other hand, refers to the demonstrated ability of individual managers to identify, assimilate and utilize new knowledge emanating from external sources. Our study argues for the linkage between PM and IACAP based on the conflicts encountered in the IACAP process.

A few studies point to the necessity for managers to manage various conflicts, especially during the assimilation and transformation stages of IACAP. For example, Enkel *et al.* (2017) suggest that in case externally derived knowledge conflicts with established knowledge and working modes, managers with ACAP must constructively adapt to such conflicts. Rafique *et al.* (2018) suggest that managers face conflicts within the ACAP process due to the overlap of roles and resource scarcity. Thus, conflicts and tensions are common between PM, IA and IACAP. In the assimilation and transformation stages, managers need to utilize their motivation and cognition to identify potentially valuable opportunities through peer discussion and persuasion leading to novel combinations with existing knowledge (Enkel *et al.*, 2017; Hafkesbrink and Schroll, 2014). PM can help manage the conflicts emanating from the tensions between old and new knowledge bases, the multiplicity of approaches suggested by different peers and the variety of decision options available to the manager. Since the activities underlying IACAP (recognition, assimilation, transformation and exploitation) require disparate skill sets and cognitive capacities (for example, activities related to recognition and transformation) (Filippini *et al.*, 2012), including multiple conflicts, PM can boost IACAP by helping manage conflicts arising out of these differences.

Hence, we hypothesize as follows:

H3. The PM of a manager enhances his or her IACAP.

H4. The IACAP of a manager mediates the relationship between PM and IA.

2.7 Learning goal orientation (LGO)

LGO refers to individuals' disposition to increase their competence and understand and further master something (Yao and Chang, 2017). LGO motivates individuals and organizations toward learning as well as creativity (Dweck, 1986; Hirst *et al.*, 2011; Islam *et al.*, 2021; Paparoidamis, 2005). Individuals with LGO believe that an individual's ability to learn and hence his/her intelligence is incremental in nature (Dweck, 1986). Such individuals are likely to take risks and seek challenges to master new knowledge, attribute any failures to lack of effort or less-than-optimal task strategies, remain stress-free despite facing setbacks and thus keep persisting with the task taken up them (Yao and Chang, 2017). They are more likely to engage and invest in learning activities and ask for feedback (Payne *et al.*, 2007).

2.8 LGO and IA

As argued earlier, ambidexterity theory has its roots in organizational learning literature. Seminal papers in ambidexterity have conceptualized the conflict between exploration and exploitation based on the adaptive learning processes and ensuing knowledge levels of organizations/individuals (Lee and Huang, 2012; March, 1991; Mom *et al.*, 2007). On the other hand, LGO is a disposition/tendency which motivates individuals toward learning, creativity and innovation. Various extant studies have highlighted a relationship between LGO and IA (Chen and Liu, 2019; Kauppila and Tempelaar, 2016; Lee and Huang, 2012; Torres *et al.*, 2015).

At the firm level, LGO has been shown to have a positive impact on ambidexterity (Chen and Liu, 2019). This is so because members operating in learning goal-oriented cultures value skill development, open-mindedness and risk-taking for the sustained success of the firms. These attributes help the firm acquire the necessary knowledge in open innovation contexts. At the individual level, scholars argue that cognitive processes that increase the depth and breadth of learning enhance innovation and creativity (Bledow *et al.*, 2009; Eisenhardt *et al.*, 2010). Recent studies have found positive impacts of LGO on IA (Kauppila and Tempelaar, 2016; Yu *et al.*, 2015). Individuals with high LGO are likely to seek challenging tasks and persist while dealing with challenging situations (Yu *et al.*, 2015). Further, individuals' tendencies to pursue learning goals instead of a singular focus on performance motivate employees to tackle exploratory activities like "search, variation, risk-taking, experimentation, play, flexibility, discovery, innovation" (March, 1991) while also looking for refinement in their day-to-day exploitative activities. In the context of frontline service employees (who are boundary spanners), prior research (Yen *et al.*, 2021; Yu *et al.*, 2015) finds that LGO first influences their engagement in cross-selling activities, and such engagements are critical for IA (service-sales ambidexterity in this case). As an indicator of the desire to take risks and maximize learning, LGO of top decision-makers (such as CEOs) enhances ambidexterity (Mammassis and Kostopoulos, 2019; Pryor *et al.*, 2021). To examine both internal and external factors influencing IA using data from executives in China, Xiang *et al.* (2019) identify a significant and positive relationship between LGO and IA. Hence, we posit as follows:

H5. The LGO of a manager enhances his/her IA.

2.9 LGO and IACAP

As discussed earlier, LGO motivates individual employees/managers toward learning, creativity and innovation. Absorptive capacity theory revolves around the capability of individuals/organizations leading to learning and assimilation of external knowledge. Hence, many scholars argue for a linkage between the two constructs (LGO and IACAP).

Prior research has found a linkage between LGO and IACAP, especially the dimensions of knowledge acquisition and assimilation (Annosi *et al.*, 2020; Chuang *et al.*, 2016; Fong *et al.*, 2018; Gong *et al.*, 2009; Hirst *et al.*, 2011; Sjödin *et al.*, 2019; Tian and Soo, 2018; Yildiz *et al.*, 2021). As individuals with high LGO believe that learning is incremental in nature, they are likely to better and persistently respond to difficulties and stresses in learning and new knowledge absorption (Yu *et al.*, 2015).

Yao and Chang (2017) argue that LGO positively impacts "individual employees' engagement in knowledge acquisition and assimilation" (p. 2044). Individuals with high levels of LGO are likely to seek new challenges and master new knowledge (Dweck, 1986); they are likely to expose themselves to external knowledge and their fit with organizational requirements. Additionally, LGO has been studied as an important necessity for building individual knowledge bases (Yao and Chang, 2017). Extant research in ACAP has found that existing knowledge bases of individuals serve as antecedents for their ability to acquire and assimilate new knowledge and hence their ACAP (Cohen and Levinthal, 1990). In the context of joint ICT project teams in Malaysia, Ojo and Raman (2015, 2017) find that LGO is a key antecedent of IACAP along with prior knowledge and the need for cognition. Similarly, examining joint project teams in the Nigerian upstream oil industry, Ojo *et al.* (2016) find a positive relationship between LGO and the knowledge assimilation dimension of IACAP. In the earlier sections, we have hypothesized how LGO is an important antecedent of both IACAP and IA. Also we have argued that IACAP acts as an antecedent of IA. We thus argue that IACAP plays a mediating role in the LGO-IA relationship.

Based on these arguments, we hypothesize as follows:

H6. The LGO of a manager enhances his/her IACAP.

H7. The ACAP of a manager mediates the relationship between their LGO and IA.

2.10 Research model

Based on the hypotheses formulated above, Figure 1 depicts the research model that will be used for empirical testing using survey data from practicing managers.

3. Methodology

We adopted a quantitative, hypothetico-deductive research approach in this paper, considering the nature of the research questions (i.e. "what" questions), the objective of generalizing from a sample to a population (of boundary spanners in this specific case) and the fact that we are looking to examine novel relationships between constructs that are already present in existing literature (Creswell and Creswell, 2017).

3.1 Sample and data collection

We conducted the study in a large Indian automotive equipment manufacturing firm (number of employees: 9,245, managerial and superintending staff: 3,673, Total revenues: INR (Indian Rupees) 120,850 million, R&D (Research and Development) intensity: 2.6%; all values as of 2019). The Indian automotive equipment manufacturing industry experiences a rapid pace of change and heightened competition, making it appropriate for studying how managers cope with the need for innovative products and a plethora of relevant, new external

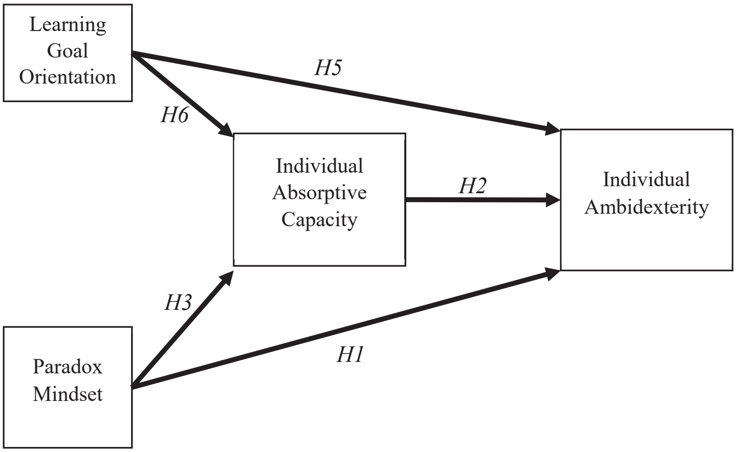

Figure 1.
Research model

knowledge. The choice of one large firm was made to keep the organizational conditions as constant as we were concerned with individual-level differences (Lowik *et al.*, 2017). Four large offices of the firm, two in north India, one in Western India and one in south India were targeted for eliciting responses from managers. We targeted technology managers who act as boundary spanners and were directly involved in activities related to R&D, product development and management, marketing and quality assurance departments of the firm.

To collect the survey data for our study, we randomly sampled 250 technology managers and sent them a personalized email containing the link to an online questionnaire between September 2018 and January 2019. We sent them three rounds of reminders, once every month. By the end of April 2019, 127 responses were returned, resulting in a 50.8% response rate. The initial part of the questionnaire asked each respondent whether they were involved in boundary-spanning activities in the last three years. We eliminated 14 responses from further analysis because of the absence of boundary-spanning roles or missing data. Hence, the final sample used for analysis consisted of 113 responses.

3.2 Measures

We used validated scales already available in extant literature to measure the constructs used in this study. We pre-tested the questionnaire with a pilot sample of 15 respondents drawn from the sampling frame to ensure there was no confusion or ambiguity concerning the survey questionnaire. Based on their feedback, we made one minor adjustment to the scale items to arrive at the survey questionnaire used for the data collection. In the scales for managerial exploration and exploitation, we slightly modified the text appearing before the items from "To what extent did you, last year, engage in work-related activities that can be characterized as follows:" to "In the last year, to what extent did you engage in:"

A seven-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree) was used to measure the items. The choice of using a seven-point Likert scale was made due to a number of reasons – the ability to capture more variance in the data optimally, achieving a balance between increase in response time and avoiding uncertain responses and less skewness in data compared to five-point scales (Dawes, 2008; DeVellis, 2016; Matell and Jacoby, 1971, 1972).

In sub-sections 3.2.1 to 3.2.5 below, we discuss the operationalization of the dependent, mediating and independent variables in more detail. Further, Table 1 describes the details of the survey measures including the sources, the item details and the factor loadings.

Item description	Factor loading
Paradox mindset (Miron-Spektor *et al.*, 2018)	
PM_2: I am comfortable dealing with conflicting demands at the same time	0.632
PM_3: Accepting contradictions is essential for my success	0.700
PM_4: Tension between ideas energizes me	0.684
PM_5: I enjoy it when I manage to pursue contradictory goals	0.728
Learning goal orientation (Yao and Chang, 2017)	
LGO_1: I am willing to select a challenging work assignment that I can learn a lot from	0.846
LGO_2: I often look for opportunities to develop new skills and knowledge	0.874
LGO_3: I enjoy challenging and difficult tasks at work where I will learn new skills	0.902
LGO_4: For me, development of my work ability in important enough to take risks	0.865
LGO_5: I prefer to work in situations that require a high level of ability and talent	0.789
Individual absorptive capacity (Lowik *et al.*, 2017)	
IACAP_1: I am always actively looking for new knowledge for my work	0.643
IACAP_2: I intentionally search for knowledge in many different domains to look 'outside the box'	0.653
IACAP_5: I frequently share my knowledge with colleagues to establish a common understanding	0.759
IACAP_6: I translate new knowledge in such a way that my colleagues understand what I mean	0.824
IACAP_7: I communicate newly acquired knowledge that might be of interest for our unit	0.724
IACAP_8: I often sit together with colleagues to come up with good ideas	0.713
IACAP_9: I attend meetings with people from different departments to come up with new ideas	0.671
IACAP_10: I develop new insights from knowledge that is available within our firm	0.680
IACAP_12: I often apply newly acquired knowledge to my work	0.681
IACAP_14: I constantly consider how I can apply new knowledge to improve my work	0.794
Manager's exploration activities – adapted from Mom *et al.* (2007, 2009)	
"In the last year, to what extent did you engage in . . ."	
EXPLORE_1: Searching for new possibilities with respect to products/services, processes or markets	0.763
EXPLORE_2: Evaluating diverse options with respect to products/services, processes or markets	0.733
EXPLORE_3: Focusing on strong renewal of products/services or markets	0.775
EXPLORE_4: Activities requiring quite some adaptability of you	0.761
EXPLORE_5: Activities requiring you to learn new skills or knowledge	0.643
Manager's exploitation activities – adapted from Mom *et al.* (2007, 2009)	
"In the last year, to what extent did you engage in . . ."	
EXPLOIT_1: Activities of which a lot of experience has been accumulated by yourself	0.757
EXPLOIT_2: Activities which serve existing (internal) customers with existing services/products	0.704
EXPLOIT_3: Activities of which it is clear to you how to conduct them	0.802
EXPLOIT_4: Activities primarily focused on achieving short-term goals	0.723
EXPLOIT_5: Activities which you can properly conduct by using your present knowledge	0.747

Table 1.
Construct details and factor loadings

3.2.1 Individual ambidexterity (IA) (dependent variable). We assessed IA using the procedures and scales developed and validated by Mom *et al.* (2007, 2009). We first measured exploration and exploitation separately using different seven-item orthogonal scales (Mom *et al.*, 2007) and then operationalized IA as the multiplicative interaction of the two, following prior research (Kauppila and Tempelaar, 2016). This multiplicative operationalization is also consistent with how ambidexterity is conceptualized theoretically (Tushman and O'Reilly, 1996).

3.2.2 Individual absorptive capacity (IACAP) (mediating variable). We assessed IACAP using the existing validated scale developed by (Lowik *et al.*, 2016, 2017), which comprises the four dimensions of IACAP – recognition, assimilation, transformation and exploitation. An exemplar scale item is "I intentionally search for knowledge in many different domains to look 'outside the box." We found the scale to be reliable (Chronbach's alpha (CA) = 0.89 and composite reliability (CR) = 0.91).

3.2.3 Paradox mindset (PM) (independent variable). We assessed PM using the scale developed and validated by Miron-Spektor *et al.* (2018). An exemplar scale item is "I am comfortable dealing with conflicting demands at the same time." We found the scale to be reliable (CA = 0.73 and CR = 0.82).

3.2.4 Learning goal orientation (LGO) (independent variable). We assessed LGO using the scale developed and validated by Yao and Chang (2017). An exemplar scale item is "I often look for opportunities to develop new skills and knowledge." We found the scale to be reliable (CA = 0.91 and CA = 0.93).

3.2.5 Control variables. Based on our review of the existing studies that use IA as the dependent variable (Kauppila and Tempelaar, 2016; Majhi *et al.*, 2020; Mom *et al.*, 2007, 2009), we controlled for the following variables in our model – age, gender, years of experience, the highest level of education and decision-making authority. Decision-making authority was measured using the scale developed and validated by Mom *et al.* (2009), based on the prior work of Dewar *et al.* (1980). An exemplar item from the scale is "I can undertake little action until my supervisor approves a decision."

3.3 Analytical approach

We used partial least squares structural equation modeling (PLS-SEM) to analyze the collected data. We opted for the use of structural equation modeling (SEM) in this study because we wanted to simultaneously test and estimate statistical relationships among multiple independent and dependent latent variables (Benitez *et al.*, 2020; Chin, 1998; Dash and Paul, 2021; Hair *et al.*, 2016; Hoyle, 1995). SEM is considered the second-generation of multivariate analysis and represents an improvement over traditional statistical techniques for the nature of the model that we are testing in this study. It allows us to answer inter-related research questions through the use of only a single, systematic and comprehensive analysis. SEM techniques are broadly classified into two categories as follows: (1) covariance based – SEM (CB-SEM) and (2) PLS-SEM.

The choice of PLS-SEM over the more traditional approach of CB-SEM was driven by recent methodological recommendations (Benitez *et al.*, 2020; Dash and Paul, 2021; Hair *et al.*, 2016):

(1) Since multiple relationships are being examined simultaneously in the form of a complex model, structural modeling is an effective technique;

(2) The data were found to be non-normal (after conducting the Kolmogorov–Smirnov and Shapiro–Wilk tests);

(3) The size of the sample (113 after removing unsuitable responses) was relatively small and

(4) The study was exploratory in nature – since we were examining the significance of hitherto unexamined relationships between constructs.

SmartPLS 3.3.3 was used to review the statistical significance of items, hypotheses, effect sizes and parameters related to the measurement model, using the recommended number of 5,000 resamples and the bias-corrected and accelerated (BCA) bootstrapping option (Hair *et al.*, 2016). To further test the model quality, we examined the values of R^2, adjusted R^2 and Q^2. While R^2 and adjusted R^2 are traditional and well-known indicators of model quality, Stone–Geisser's Q^2 (Geisser, 1974; Stone, 1974) evaluates the cross-validated predictive relevance of the PLS model with value thresholds of 0.02. 0.15 and 0.35 representing small, medium and large effects, respectively (Hair *et al.*, 2016).

We used the bootstrapping approach (Preacher and Hayes, 2008) to test the mediation effects, since recent research (Hayes, 2018; MacKinnon *et al.*, 2007; Zhao *et al.*, 2010) recommends it over traditional approaches such as the joint significance test and the Sobel test. Bootstrapping procedure involves non-parametric resampling and does not assume a normal distribution for the sampling distribution. It is a computationally intensive method involving repeated sampling from the data (with replacement) and the estimation of the indirect effect for each of these repeated samples. Finally, by repetition of this process thousands of times, the sampling distribution of the indirect effect is estimated and confidence intervals are constructed (Preacher and Hayes, 2008). We used the PROCESS MACRO with IBM SPSS (Statistical Package for the Social Sciences) Statistics v23 to conduct the mediation analysis using Model 4 and the recommended 5,000 resamples. In the bootstrapping approach, the mediation effect is considered to be significant if the 95% confidence interval of the bootstrapped mediation effect data does not contain zero. Also we checked the mediation effect size using percent mediation (P_M) and completely standardized indirect effect (ab_{cs}) (Preacher and Kelley, 2011).

Finally, as a post-hoc test for the mediation analysis, we checked for the potential issue of reverse causality as recommended in prior research (Landis and Dunlap, 2000; Stone-Romero and Rosopa, 2008). We exchanged the independent and dependent variables and found that the mediation models broke. This suggests that reverse causality is not a concern and the mediation model proposed by us is robust.

4. Findings and analysis

In this section, we report the findings obtained from the empirical validation of our research model. Following accepted conventions in the use of PLS-SEM (Cepeda-Carrion *et al.*, 2019; Hair *et al.*, 2016), the following sections describe (1) the checking for common-method and non-response biases (NRBs), (2) the assessment of the measurement and structural models and then (3) the mediation analysis results.

4.1 Checking for common-method and non-response biases

Common-method bias (CMB) occurs due to the collection of data using a single instrument and means that variance could potentially be attributable to the measurement method rather than the construct (Podsakoff *et al.*, 2003). We tried to minimize CMB during the survey design using accepted methodological recommendations (Podsakoff *et al.*, 2003). Post data collection, we used Harman's single factor test (Harman, 1967) and found that the single unrotated factor accounted for only 29.4% of the variance. Thus, we can safely assume that CMB is not a serious concern in this study.

We used the wave method (Armstrong and Overton, 1977) to check for NRB. Comparing the early (first 30) and late (last 30) respondents using *t*-tests, we found no significant differences ($p > 0.1$) for any of the indicators, including those related to respondent demographics. Thus, we can conclude that NRB is also not a serious concern for this study.

4.2 Assessment of the measurement model

The reflective measurement model was assessed by evaluating (1) internal consistency, (2) convergent validity and (3) discriminant validity.

We assessed internal consistency using CA and CR, following recommendations that the former underestimates the reliability and the latter overestimates the same (Hair *et al.*, 2016). Table 2 reports the results of assessing the measurement model. For all the constructs, the CA and CR scores were above the recommended threshold value of 0.7.

Regarding convergent validity, the accepted convention is to have average variance extracted (AVE) over 0.50 and outer loading values above 0.70. The AVE for all the constructs was greater than 0.50. We retained those items whose outer loading values were between 0.6 and 0.7, as this range is considered acceptable in social science research (Hair *et al.*, 2016). Thus, we retained items with loadings above 0.6 if removing them did not enhance the CR or AVE (See Table 1 for factor loadings).

We established the discriminant validity by examining the cross-loadings and using the Fornell–Larcker criteria. No indicator had a higher loading on a construct other than its own. Also for each construct, the square root of AVE was greater than the inter-construct correlations (See Table 2). Finally, we looked at the heterotrait-to-monotrait (HTMT) ratios to also confirm the discriminant validity.

4.3 Assessment of the structural model

First, we assessed the variance inflation factor (VIF) values which were all found to be less than 3.3. Thus, multicollinearity is not an issue in this study (Hair *et al.*, 2016).

Then, we used the standardized root mean square residual (SRMR) value to assess overall model fit and thus validate the model. The acceptable SRMR threshold values for CB-SEM and PLS-SEM are 0.08 and 0.10, respectively (Hair *et al.*, 2016). Estimation of our model using the PLS algorithm yielded an SRMR value of 0.089, which is acceptable (≤ 0.10) and exhibits the goodness of fit of our model.

We next checked the model quality indices such as R^2, adjusted R^2 and Q^2. All these indices are satisfactory for our study, and we report the details in Table 3. The Geisser–Stone Q^2 values indicate medium and strong predictive relevance for individual ACAP and IA respectively.

Table 4 shows the results of the testing of the proposed hypotheses. Except H5, all the other hypothesized relationships were found to be statistically significant for our sample.

	Mean	SD	AVE	CA	CR	LGO	PM	IAC	IA
LGO	5.87	0.94	0.73	0.909	0.932	*0.856*			
PM	4.99	0.87	0.56	0.730	0.822	0.452	*0.693*		
IAC	5.56	0.91	0.51	0.893	0.913	0.682	0.513	*0.716*	
IA	26.46	5.43	–	–	–	0.536	0.566	0.683	*1*

Table 2. Descriptive statistics, consistency, reliability and divergent validity

Model quality index	Individual ACAP	Individual ambidexterity
R^2	0.518 ($t = 8.992, p = 0$)	0.533 ($t = 9.023, p = 0$)
Adjusted R^2	0.509 ($t = 8.682, p = 0$)	0.520 ($t = 8.570, p = 0$)
Q^2	0.254 (medium effect)	0.508 (strong effect)

Table 3. Model quality indices

Hypothesis	Path	Coefficient	*T*-statistic	*p*-value	Result
H1	PM → IA	0.283***	3.705	0.000	Supported
H2	IAC → IA	0.485***	4.835	0.000	Supported
H3	PM → IAC	0.257***	3.641	0.000	Supported
H5	LGO → IA	0.078	0.857	0.391	Not supported
H6	LGO → IAC	0.566***	7.971	0.000	Supported

Note(s): *** significant at 95% confidence level

Table 4. Results of hypothesis testing

Finally, Table 5 describes the results of the mediation analysis. Both the mediation hypotheses found statistical support in our sample. The results indicate that while IACAP partially mediates the relationship between PM and IA, it fully mediates the relationship between LGO and IA.

5. Discussion and conclusion

Absorptive capacity and ambidexterity are important knowledge-related capabilities and their positive impacts on a variety of outcomes are well examined in prior academic literature (Lichtenthaler, 2016; Mom *et al.*, 2019). While scholars have begun examining IA as well as IACAP (Enkel *et al.*, 2017), confusion about the similarity or the lack of it between these two concepts still persists at the individual level of analysis. Further, despite a lot of academic attention on the antecedents of IA and IACAP, there is a lack of investigation of a key learning-related antecedent (LGO) and conflict management-related antecedent (PM). The objective of this paper was to examine LGO and PM as the antecedents of IACAP and IA and further see if IACAP mediates the relationship between these two antecedents and IA.

The study reveals important interactions between managerial mindsets and knowledge-based capabilities. PLS-SEM analysis demonstrates the mediating effect of IACAP between PM and IA. LGO is also found to positively impact IA via the mediation of IACAP. While IACAP partially mediates the relationship between PM and IA, there is full mediation in the case of LGO. We find that IACAP plays an important mediating role for both PM's and LGO's impact on IA. Next, we lay out our study's theoretical contributions and managerial implications.

5.1 Theoretical contributions

Our study argues the importance of persistent conflicts and their management for both IA and IACAP for boundary-spanning managers. Recent research has established conflicts as the micro-foundational basis of ambidexterity (Martin *et al.*, 2019). Similarly, conflicts need to be dealt with during the assimilation and transformation stages of ACAP (Jansen *et al.*, 2005; Rafique *et al.*, 2018). New research has also found that managers who display PM, i.e. who are accepting and motivated by persistent and interrelated conflicts can display better innovative work behavior and on-role job performance (Miron-Spektor *et al.*, 2018). Our study explores the impact of PM on both IA and IACAP and finds two distinct impacts on IA – direct as well mediated through IACAP.

Another similarity between IA and IACAP is their base in managerial and organizational learning (March, 1991). While the concept of ACAP entails the recognition, assimilation, transformation and implementation of external knowledge, the concept of ambidexterity focuses on how entities can balance exploratory and exploitative learning. Our study finds that LGO impacts both IACAP and IA. However, we also find that IACAP fully mediates the relationship between LGO and IA. This means that if a boundary-spanning manager requires

Table 5.
Mediation analysis results

H#	IV → M → DV	DE (c')	IE (a*b)	TE (c)	95% CI of, i.e.	ab_{cs} (95% CI)	P_M	Result
H4	PM → IAC → IA	0.2934	0.2730	0.5664	[0.1686, 0.4085]	[0.1825, 0.3761]	0.4821	Partial mediation
H7	LGO → IAC → IA	0.1314	0.4048	0.5363	[0.2464, 0.6132]	[0.2632, 0.5410]	0.7549	Full mediation

Note(s): IV = independent variable, M = mediating variable, DV = dependent variable, DE = direct effect, IE = indirect effect, TE = total effect, CI = confidence interval, ab_{cs} = completely standardized indirect effect and P_M = percent mediation an LGO, he/she can recognize, assimilate and implement new knowledge. The manifestation of such ACAP then in turn, promotes ambidexterity in the manager's behavior.

These two sets of findings greatly enhance our understanding of the inter-relationship between IACAP and IA. We confirm that a PM and conflict management are important antecedents of IA. IACAP mediates the relationship of both PM and LGO with IA. Our findings help in clearly distinguishing the role of two linked and important learning and innovation-based constructs of ambidexterity and ACAP.

5.2 Managerial implications

Our study has important implications for boundary-spanning employees in organizations. Increasingly, boundary spanners are being asked by their top management to identify and implement innovative technologies, systems, processes and knowledge in their firms (Griffin *et al.*, 2007; Kauppila and Tempelaar, 2016). Boundary-spanning managers, tasked with interfacing with external sources of knowledge, must be able to display the ability to absorb and implement new knowledge while at the same time balancing between exploratory and exploitative learning. These two capabilities must be displayed in tandem to produce sustainable innovative work behavior.

Further, PM and LGOs are found to be important antecedents that help in promoting both capabilities. To bolster IA, individual managers must adopt a risk-taking and challenge-seeking approach toward new knowledge. Further, the managers must also be comfortable with the ensuing conflicts while pursuing IA and ACAP.

From the top management perspective, our study has implications regarding the traits required for boundary-spanning job descriptions. While recruiting managers for ambidextrous boundary-spanning roles, the top management should prioritize the presence of LGO and paradoxical mindset in candidates. To facilitate an organizational culture promoting IA, organizations must promote avenues for knowledge dissemination, especially for new external knowledge, while accepting the inevitable conflicts that employees must face in the pursuit of ambidexterity.

5.3 Limitations and future research

This paper is not free from its limitations. The nature of relationships hypothesized in the study is based on extant academic literature and its analysis. However, because of its cross-sectional nature, this study has its limitations as far as testing the causal nature of relationships is concerned despite the common method and NRB shown as not being significant concerns. Also it is not possible to make causal inferences about the mediation effects because of the cross-sectional data collection (Stone-Romero and Rosopa, 2008, 2010). Experimental and longitudinal studies may be conducted to address this limitation.

This paper utilizes a quantitative study in a single firm with the objective to examine the heterogeneity encountered at an individual level instead of inter-firm differences. While single case studies provide novel conceptual insights, further research is required to test the inter-firm generalizability of our findings by testing the model in varied contexts.

Our study focuses exclusively on boundary spanners since the extant literature notes that the conflicts they confront make the absorption and implementation of new knowledge a critical activity. Future research should test our findings in non-boundary-spanning contexts to find out the generalizability of our theoretical arguments.

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About the authors

Saurav Snehvrat is Assistant Professor in the Strategic Management Area at XLRI-Xavier School of Management, Jamshedpur, India. His research interests include paradox theory, organizational ambidexterity and the base of the pyramid (BoP). Saurav Snehvrat is the corresponding author and can be contacted at: saurav@xlri.ac.in

Sanjay Chaudhary is Associate Professor in the area of Strategic Management. He holds Fellow in Strategic Management from Indian Institute of Management, Kashipur. His research interests include knowledge management and entrepreneurial orientation.

Siddharth Gaurav Majhi is Assistant Professor in the area of Information Systems and Technology. He holds a PhD in Information Systems from Indian Institute of Management, Ranchi. His research interests include dynamic capability, ambidexterity and information systems adoption.